Article 5 : A New Direction

1. Introduction

a. Journey Synthesis

This article marks a turning point in the Bitcoin price prediction project documented by a series of articles.

As a reminder, the approach presented was articulated as follows:

- Article 1: Establishing the methodological framework and the tested hypothesis: "Is it possible to predict future prices from past prices alone?"
- Article 2: Building the foundations. Creation of a data pipeline (Binance API), and Exploratory Data Analysis (EDA). Key results: fragmented market, retail dominance, and especially frequent regime breaks.
- Article 3: Core of the experimentation. ML Modeling (LSTM, GRU, SimpleRNN) approaching real conditions (walk-forward). Conclusion: The initial hypothesis is invalidated.

b. Discussion: Why the Time Series Approach Reaches Its Limits

It emerges from this experimentation that the univariate prediction approach failed, not due to technical errors, but due to structural limitations.

The cross-analysis of Articles 2 and 3 reveals three blocking points:

- Signal-to-Noise Ratio too low: Models converge towards means or repeat initial patterns. This indicates that the price curve is dominated by random noise (random walk) rather than deterministic patterns exploitable by a neural network on this time window.
- Structural Non-Stationarity: An ML model assumes that the future will statistically resemble the past. However, the EDA (Art 2) showed that the crypto market changes regime too quickly (semester cycles valid in 2022, are obsolete in 2025). A trained model can become useless overnight.
- Platform vs Price Disconnection: Buyer/seller pressure on Binance is balanced, yet the global price rises. Movements are initiated elsewhere (institutional, OTC).

Intermediate Conclusion: Price is an output variable too dependent on external factors invisible in an isolated time series.

c. What Now? Moving to a New Project

Faced with this observation, two paths are possible:

- Complexify Price Prediction: Add external variables (sentiment, on-chain, macro) to attempt to capture missing factors. This is a valid approach, well documented in quantitative literature.
- Change the Problem: If the "When" (timing) is unpredictable or out of reach, then the "What" (asset selection) becomes the critical performance variable.

It is towards this second path that I will orient the continuation of my actions.

The objective of this project has always been to explore the cryptoverse with a data compass. Since price is the noisy result of a multitude of factors, let us go back to the source. Instead of predicting the curve, let us evaluate the solidity of what generates it: the project itself.

A Positioning Choice

Research on price prediction by time series is an interesting field, but highly competitive. As a solo project, my added value does not lie in the race to optimize models on such a noisy signal. My expertise and interest lie in the systemic understanding of projects, reliable data engineering, and the construction of decision support tools.

Abandoning the price approach is therefore not only an empirical conclusion: it is also a conscious arbitrage. Rather than seeking to predict when the market moves, I choose to focus where my added value lies: a problem of evaluation and classification, more aligned with my skills.

Towards Assisted Investment

In an ecosystem where scams are becoming professionalized and where technologies evolve rapidly (DePIN, RWA, AI), the main risk is no longer only volatility, but also reliability.

This article therefore proposes a pivot: moving from an Assisted Trading logic (price prediction) to an Assisted Investment logic (fundamentals scoring).

We will define how to use the digital traces left by projects to build a risk classification system. Unlike price, these traces are input variables that are more stable and observable.

The rest of this document will detail the architecture of this scoring system, paving the way for a series of deep dive articles devoted to the extraction and analysis of each of these metrics.

2. The Traces

Unlike traditional projects (Web2.0), blockchain projects possess structural specificities inherited from Bitcoin, which make them particularly conducive to data analysis:

- **Transparency:**
  - **On chain:** All transactions are public and verifiable on the blockchain.
  - **Technical (Open Source):** Most codes are public (e.g., GitHub).
  - **Roadmap (Whitepaper):** Objectives are documented. Although subject to interpretation, this text constitutes a semantic basis analyzable by NLP (Natural Language Processing).
- **Decentralization:** A Web3 project cannot exist without active users.

These actions form a base of digital artifacts, but for a standard user, the analysis of these indicators is done manually and subjectively. This is where Data Science brings its added value: transforming qualitative signals into objective quantitative variables.

Unlike price (non-stationary, Art 3), these traces are input variables that are more stable. They do not serve to predict a price in the short term, but to classify the probability of survival and success of a project.

Here are the categories of metrics identified to build this scoring system, as well as the question that must be answered by each category:

- **Team:** "Who carries the project and are they credible?"
- **Technology & Code:** "Is the product functional and secure?"
- **Tokenomics & Finance:** "Is the economic structure viable?"
- **Documentation:** "Is the vision clear and coherent?"
- **Community:** "Is adoption real or artificial?"

These categories set the framework for the analysis. It remains now to define the method to transform them into quantifiable and usable metrics.

3. The Roadmap

As explained in Article 2, any artificial intelligence project relies on data quality. This is the first challenge to overcome. For Bitcoin, price data is abundant and structured. For project evaluation, the situation is more complex: there is no reference labeled dataset allowing to objectively distinguish a "reliable" project from a "risky" project.

Unlike price prediction where ground truth is known *a posteriori* (the future price), evaluating the reliability of a project is intrinsically subjective. A project may seem solid and fail (bad timing, macro) or be a sophisticated scam and prosper temporarily.

Faced with this absence of ground truth, a Big Bang approach (building everything at once) would be doomed to failure. I therefore prefer an iterative approach, inspired by CI/CD principles (Continuous Integration / Continuous Deployment), adapted to Data Science:

a. Metrics Deep-Dives Metrics (Research & Validation)

Each pillar identified in Section 2 will be the subject of dedicated exploration:

- Bibliographical Research: Accumulation of diverse sources (scientific articles, industry reports, ...) to identify validated indicators.
- Publication: Each Deep-Dive will result in an article summarizing the insights and the selected metrics.

b. Tool Development (Iterative)

For each metric, development will follow a three-step cycle:

- **Source Identification:** GitHub API, Etherscan, social networks, fundraising databases...
- **POC (Proof of Concept):** A simplified version to validate technical feasibility and data access.
- **Progressive Enrichment:** Addition of functionalities (cleaning, normalization, scoring) as feedback comes in.

c. Dataset Construction (Hybrid Labeling)

- Phase 1 (Manual): Writing 100% manual project analyses to establish an initial ground truth base and validate the relevance of chosen metrics.
- Phase 2 (Assisted): Progressive integration of automation tools with systematic human validation to limit biases.
- Phase 3 (Evolving Dataset): As the base grows, it will serve to train classification models, whose performance will be compared to manual analyses.

d. Transparency and External Validation

- Open-Source Code: All extraction and scoring scripts will be public for auditability.

- Feedback Loop: Regular publication aims to create a form of open peer review, allowing validation of chosen metrics before their integration into the global system.
- Continuous Documentation: Each validated step will be documented (articles, notebooks, commented code).

Next Step: Article 6

With this roadmap launched, place to the first concrete implementation. The next article will be a Deep-Dive on the Whitepaper, an element that I find fascinating because it crystallizes both the technical, economic, and marketing vision of a project. We will explore there how to transform this qualitative document into quantitative metrics exploitable via NLP.